UGG: Unified Generative Geometry Model

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Abstract

Understanding 4D scenes is a core challenge in visual intelligence, requiring models to jointly capture scene geometry, appearance, and dynamics from partial observations. Existing approaches, however, are subdivided into feedforward models that are used for direct geometric prediction and generative models for visual synthesis. We propose a unified generative-predictive framework for 4D scene understanding that jointly models RGB video and geometric representations such as depth and camera pose with a single joint generative model. Our generative formulation enables us to learn from diverse, heterogeneous datasets with different subsets of geometric and visual annotations, and enables us to condition on arbitrary subsets of inputs, allowing a single model to perform cross-modal generation, geometric prediction, and future scene inference from partial observations. In addition, our generative approach enables test-time search over multiple candidate scene completions, improving geometric prediction from sparse observations. Overall, we illustrate the efficacy of our generative-predictive framework on a suite of generation and prediction tasks.

1 Introduction

For embodied agents to perceive and act intelligently in the physical world, they must understand its 3D structure and predict how it will evolve over time [10, 52, 3]. However, visual observations provide only partial 2D projections of the world, each capturing a limited set of viewpoints, visible surfaces, and changes over time. Robust 4D scene understanding therefore requires inferring what is not directly observed, namely the 3D geometry, camera viewpoint, scene appearance from unseen views, and scene evolution.

Current methods have made strong progress on different parts of this problem, but they are often separated into geometric prediction and visual generation. Feedforward models can directly estimate scene geometry, including depth, camera pose, and 3D structure, from images or videos. These methods are efficient and often accurate under the training distribution, but they typically produce a single deterministic estimate and can struggle when the observations are sparse or out of distribution. Generative models, in contrast, learn strong visual priors for completing missing observations, synthesizing novel views, and predicting plausible dynamic futures. However, they often lack explicit geometric grounding, which can lead to outputs that are visually plausible but inconsistent in 3D. Recent work has begun to narrow this gap by repurposing generative video models for geometric prediction [19, 28, 68], or by injecting geometric representations into generative models to improve 3D consistency [31, 48, 18]. Nevertheless, generation and prediction are still commonly treated as separate capabilities, rather than as two forms of inference in the same underlying scene model.

In this work, we introduce UGG, a unified generative geometry model that treats visual generation and geometric prediction as different conditional queries of the same underlying 4D scene representation.

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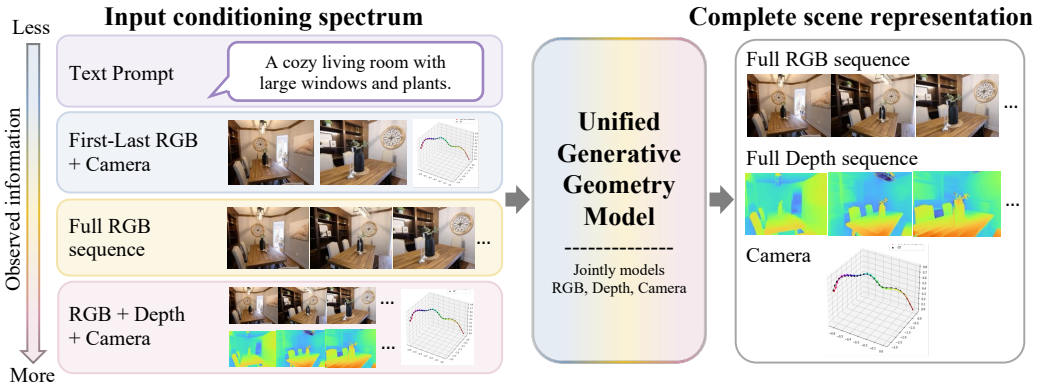


Figure 1: **A unified model for generation and prediction.** We formulate visual generation and geometric prediction as conditional completion in a single joint model over RGB, depth, and camera pose. Given arbitrary partial observations, ranging from a text prompt to RGB sequences with geometric cues, our model jointly uses these modalities to infer a complete 4D scene representation, enabling both generation from sparse inputs and geometric prediction from observations.

The appearance (RGB video), 3D structure (depth), and viewpoint (camera geometry) all provide complementary observations of a dynamic scene. Depending on what is observed, a joint model can infer missing scene components from the observed ones.

We instantiate this formulation by representing RGB, depth, and camera geometry as spatially aligned modalities, with camera pose encoded as dense ray maps. A multimodal transformer jointly processes these representation through shared self-attention and modality-specific layers. Modality flexibility is achieved through an extension of Diffusion Forcing [9]: different tasks are specified through noise patterns over both modality and time, where observed components serve as conditioning context and missing components are denoised. This conditional completion mechanism enables the same model to condition on a text prompt, a single RGB frame, an RGB sequence, camera poses, depth maps, or any combination of these, and complete the remaining modalities. Moreover, the model can learn from heterogeneous datasets by applying the loss only on available modalities, instead of requiring every example to contain complete annotations of pose and depth.

Our generative formulation also enables inference-time search over multiple plausible scene completions. Instead of committing to a single geometric prediction, we sample multiple candidates and select the one with the strongest cross-modal consistency, measured through photometric reprojection. This inference-time search capability is also consistent with the cognitive theory of perception as inference: human scene understanding requires inferring latent scene causes from incomplete observations, and may rely on internal simulation to evaluate possible scene explanations [22, 56, 2, 29].

Overall, our contributions are threefold: **i)** We formulate 4D scene understanding as joint generative modeling over appearance and geometry, treating visual generation and geometric prediction as missing-modality denoising within a single conditional model. **ii)** We introduce a unified generative-predictive model that represents RGB, depth, and camera pose as spatially aligned modalities, enabling arbitrary conditioning and learning from heterogeneous datasets with incomplete supervision. **iii)** Finally, we leverage the generative formulation for inference-time search over multiple 4D scene hypotheses, using geometric consistency to improve geometric prediction.

2 Related Work

Generative Modeling for 4D Scene Understanding. Geometric prediction is largely dominated by feedforward discriminative models that regress depth, camera pose, and scene flow, or 3D structure from visual observations [43, 23, 58, 41, 33, 50, 45, 21]. While these methods achieve strong performance, they typically produce deterministic, single-pass estimates that struggle with inherent visual ambiguities, including sparse-view observations. To address these limitations, recent work has explored generative priors for pose estimation [8, 57], depth prediction [20, 15, 32], and dynamic 4D reconstruction [19, 68]. Generative models have also been used for novel view synthesis [55, 54]. However, these methods typically treat geometric prediction as a unidirectional mapping from RGB to specific geometric outputs and rely on strictly paired datasets.

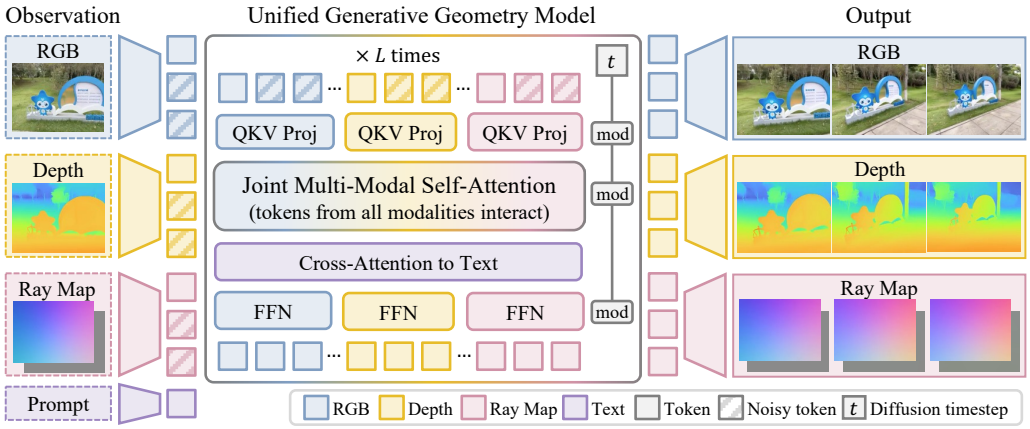


Figure 2: **Generative-predictive transformer.** We model RGB, depth, and camera pose as spatially aligned tokens, with camera pose represented as dense ray maps. Partial observations are zero-noise context tokens, while missing modalities are noisy tokens to be denoised. A unified transformer processes all tokens with modality-specific projections and feed-forward layers together with shared multimodal self-attention, enabling information exchange across modalities. The decoded RGB, depth, and ray maps form a complete 4D scene representation for cross-modal generation, geometric prediction, and inference-time consistency search.

Geometry-aware video and 4D generation. Recent generative world models have advanced from purely visual synthesis [4, 39, 1, 10] toward geometry-aware generation. One line of work injects 3D scene structure or persistent spatial memory into video generation to improve spatial consistency [31, 59, 18, 49, 64, 51]. Another line studies camera-conditioned generation, using camera trajectories to control synthesized videos [14, 63, 47]. More closely related, One4D [28] and WorldReel [13] jointly generate RGB frames and 4D geometric representations from text or RGB prompts. However, these methods typically follow fixed conditioning and output pathways. In contrast, we treat RGB, depth, and camera rays as symmetric modalities, enabling an any-to-any conditional completion framework for both generation and prediction.

Unified conditional modeling. Our work is related to flexible conditional generation, where diverse tasks are cast as missing-modality completion. Mixture-of-Transformers (MoT) architectures [24], such as Motus [3] and DreamZero [52], show that a single model can process heterogeneous modalities. For flexible conditioning, we build on Diffusion Forcing [9] and DFoT [36], which assign different noise levels to different tokens or frames. Recent works extend this paradigm for long-horizon video generation or embodied planning [11, 42]; in contrast, we apply it to multimodal 4D scene understanding, treating RGB, depth, and camera geometry as symmetric components. This probabilistic formulation also enables inference-time search. Motivated by test-time scaling [61, 27], we use additional inference compute to compare multiple candidate scene completions. We then select the candidate with stronger photometric reprojection consistency to improve geometric prediction.

3 Method

We propose a unified generative-predictive framework for 4D scene understanding. Given partial observations of a scene, our goal is to infer a complete scene representation containing appearance, depth, and camera geometry. We formulate both visual generation and geometric prediction as conditional completion in a joint generative model over spatially aligned RGB, depth, and camera-ray modalities. We specify each task by a noise pattern over multimodal tokens, with observed tokens as zero-noise context and missing tokens as noisy targets. The framework supports heterogeneous training data with incomplete modality supervision, arbitrary partial conditioning, and test-time search over multiple 4D scene hypotheses.

3.1 Unified 4D Scene Representation

We represent a 4D scene as a multimodal sequence of N frames. Each frame i contains an RGB image I_i, the depth map D_i, camera parameters $C_i = (K_i, T_i)$, where K_i denotes the camera intrinsics and T_i denotes the camera extrinsics. In practice, only a subset of this information may be observed. An example may contain one or more RGB frames, optionally with depth or camera parameters, or only

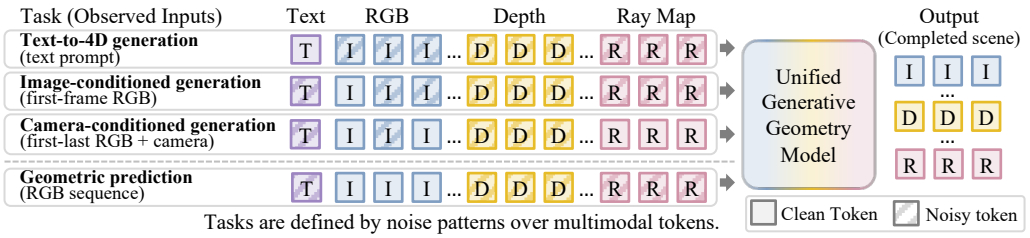


Figure 3: **Conditional completion with noise as masking.** We specify each task by the clean/noisy pattern over RGB, depth, camera-ray, and text tokens. Clean tokens provide observed context, while noisy tokens are denoised as missing targets, enabling generation and prediction with the same model architecture.

a text prompt. Our model is therefore designed to infer missing scene components from arbitrary partial observations.

To spatially align camera geometry with RGB and depth, we represent camera parameters C_i as a dense map of 6D Plücker rays, following prior ray-based scene and camera representations [35, 57]. This representation turns camera geometry into an image-like modality. Each pixel (u, v) is associated with a ray moment $\mathbf{m}_i(u, v)$ in a common world coordinate system, constructed from the viewing direction $\mathbf{d}_i(u, v)$ and the camera center $\mathbf{o}_i$.

$$R_i(u, v) = (\mathbf{d}_i(u, v), \mathbf{m}_i(u, v)), \qquad \mathbf{m}_i(u, v) = \mathbf{o}_i \times \mathbf{d}_i(u, v). \tag{1}$$

The resulting scene representation $\mathcal{S}$ contains the appearance and local 3D geometry for each frame i:

$$\mathcal{S} = \{(I_i, D_i, R_i)\}_{i=1}^{N}.$$

3.2 Generative-Predictive Transformer

Given the spatially aligned representation $\mathcal{S}$, we design a multimodal diffusion transformer that jointly processes appearance, depth, and camera-ray geometry (Fig. 2).

Aligned Modality Tokenization. RGB frames and depth maps are encoded into latent tokens using a VAE encoder, producing z^I and z^D. Before encoding, we normalize depth and camera rays using a shared scale factor, making reprojection and cross-frame geometric consistency well defined. Ray maps are downsampled to the latent spatial resolution and projected directly into ray-map tokens. Since a ray map contains both ray directions and ray moments, we represent it with two token groups $z^R = [z^{R_d}; z^{R_m}]$, where z^{R_d} denotes ray-direction tokens and z^{R_m} denotes ray-moment tokens. The complete multimodal token sequence is

$$z = [z^I; z^D; z^{R_d}; z^{R_m}].$$

Each token is processed with modality specific patch embedding layers and a spatiotemporal RoPE positional encoding over latent spatial coordinates and the frame index [38].

At the output stage, RGB and depth tokens are decoded back to pixel space using the VAE decoder. Ray map tokens are directly mapped back to dense ray maps, which can be converted back to camera parameters C_i including rotation, translation, and intrinsics.

Diffusion Mixture-of-Transformers. The backbone uses a mixture-of-transformers [24] design. Each modality has its own input projections, allowing for modality-specific representation structure. After modality-specific projection, all modalities are concatenated and processed by shared self-attention, where information can be exchanged across modalities. Modalities are then separated again to modality-specific feed-forward layers and output heads. Optional text prompts are injected through cross-attention layers.

Each token has its own diffusion timestep. These timesteps are embedded and injected into transformer blocks through adaptive layer normalization modulation [30], allowing the model to process clean context tokens and noisy target tokens at different noise levels.

3.3 Conditional Completion with Noise as Masking

We formulate generation and geometric prediction as conditional denoising over a shared multimodal representation. Let $\mathcal{M} = \{I, D, R_d, R_m\}$ denote the set of modalities, and let $z_{i,0}^m$ denote the clean

latent token at spatiotemporal position i for modality m. Which modalities are revealed or corrupted specify different tasks without changing the architecture. For example, RGB-conditioned geometric prediction reveals RGB tokens while treating depth and camera-ray tokens as prediction targets.

Task Specification by Noise Pattern. Diffusion forcing enables flexible conditional generation by assigning independently sampled noise levels across temporal positions [9]. We extend this formulation to multimodal sequences, allowing noise levels $t_i^m \in [0, 1]$ to vary across both modalities and temporal positions. We construct the noisy input $\tilde{z}_i^m$ through linear interpolation of the clean token and Gaussian noise

$$\tilde{z}_i^m = (1 - t_i^m) z_{i,0}^m + t_i^m \epsilon_i^m, \qquad \epsilon_i^m \sim \mathcal{N}(0, I). \tag{2}$$

At $t_i^m = 0$, the token is fully clean, while at $t_i^m = 1$, it is pure noise. By varying the noise levels across both modality and temporal position, we realize different conditional generation and prediction tasks without modifying the model architecture.

Training with Heterogeneous Data. During training, we sample conditioning patterns that designate subsets of multimodal tokens as context, and the remainder as prediction targets. Context tokens receive either zero or reduced noise levels, while target tokens have independently sampled noise levels. With probability p_{clean}, context noise levels are set to $t_i^m = 0$ to allow for fully clean context conditioning.

The model is trained using conditional Flow Matching [26], with target velocity $v_i^m = \partial \tilde{z}_i^m / \partial t_i^m = \epsilon_i^m - z_{i,0}^m$. Given the corrupted multimodal tokens $\tilde{z}$, noise levels t, and optional text conditioning c, the model predicts $\hat{v} = f_\theta(\tilde{z}, t, c)$, with the loss:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t,\epsilon} \left[\sum_{i,\, m \in \mathcal{M}} a_i^m \lambda^m \left\| \hat{v}_i^m - v_i^m \right\|_2^2 \right]. \tag{3}$$

λ^m is a weight balancing the losses for each modality. Modality availability is denoted as $a_i^m \in \{0, 1\}$. The availability mask allows a single model to be trained using heterogeneous mixes of RGB-only, RGB-camera, RGB-depth, and RGB-depth-camera data.

Inference with History Guidance. At inference time, the same noise pattern is used, but conditioning is stricter. Observed tokens are clamped to their clean values at every denoising step and assigned $t_i^m = 0$, while unknown tokens are initialized from Gaussian noise and progressively denoised. For example, in a first-last-frame camera-controlled setting, the first and last RGB tokens and all camera ray tokens are clamped as context, while intermediate RGB and depth tokens are generated. If depth is not provided as an input, it is not clamped and is inferred jointly with the generated RGB sequence.

We also use History Guidance [36] to strengthen consistency with the observed context. Let $\mathcal{H}$ denote the observed history context tokens, including RGB frames, ray maps, depth maps, or any combination of modalities. We compute a history-conditioned flow $\hat{v}_{\mathcal{H}}$ with tokens in $\mathcal{H}$ clamped to their observed values, and a history-masked flow $\hat{v}_{\varnothing}$ by replacing the same context tokens with random noise. The guided flow is

$$\hat{v}_{\text{HG}} = \hat{v}_{\mathcal{H}} + w \left(\hat{v}_{\mathcal{H}} - \hat{v}_{\varnothing} \right), \tag{4}$$

where w is the history-guidance strength. This is analogous to classifier-free guidance, but the dropped condition is the observed history/context rather than the text prompt.

3.4 Inference-time Search

The generative formulation also enables inference-time search. Given the same observed context, we sample L candidate completions by running the denoising process from different random initializations. Each sample yields a complete 4D scene hypothesis:

$$\hat{\mathcal{S}}^{(\ell)} = \{(\hat{I}_i^{(\ell)}, \hat{D}_i^{(\ell)}, \hat{R}_i^{(\ell)})\}_{i=1}^N, \qquad \ell = 1, \ldots, L.$$

From this set, we then choose the candidate that is most geometrically self-consistent. For each candidate, we evaluate a photometric reprojection error on a lightweight set of frame pairs, measuring whether a candidate's RGB, depth, and camera predictions are mutually consistent. For each pair (i, j), we reproject the RGB from frame i to the perspective of frame j using the predicted depth and camera geometry, and then compare it to the candidate's RGB prediction of frame j. Candidates with inconsistent geometry or appearance produce inaccurate warps and have higher error.

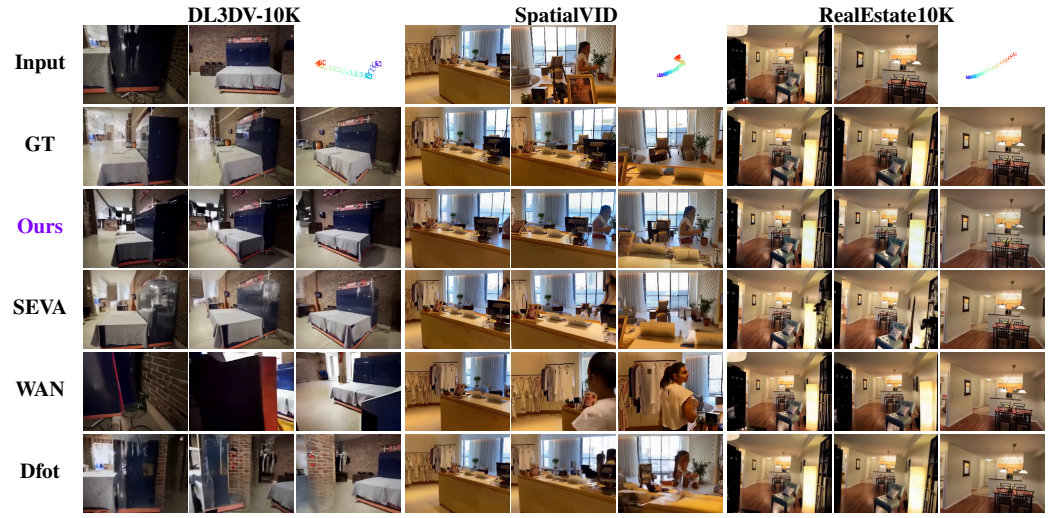


Figure 4: **Qualitative comparison of camera-controlled generation.** Given the first and last RGB frames and a target camera trajectory, we compare generated intermediate frames against ground truth and baseline video generation methods. UGG better preserves scene identity and dynamic content in compared with the generic video completion baselines.

4 Experiments

We evaluate UGG on a suite of 4D scene understanding tasks, including camera-controlled generation (Sec. 4.2 and geometric prediction. Geometric prediction tasks under both dense and sparse observation settings include camera pose estimation (Sec. 4.3) and video depth estimation (Sec. 4.4). Our single unified model handles all tasks by changing only the clean/noisy masking pattern, without requiring task-specific architectures. We further show that UGG's generative formulation enables inference-time search, improving predicted scene consistency with additional samples (Sec. 4.5). Additional visualizations are provided on the project page.

4.1 Model Training

We train on a weighted mixture of real, synthetic, and robotic video datasets, including DL3DV-10K [25], RealEstate10K [65], SpatialVID [40], ScanNet++ [53], AgiBot World [5], OmniWorld [66], Virtual-KITTI2 [7], and PointOdyssey [62]. These datasets provide heterogeneous supervision over RGB video, depth, camera pose, and text; missing modalities are omitted from the loss. UGG is trained with multiple clean/noisy mask patterns over modality tokens to cover generation and geometric prediction tasks. UGG weights are initialized from Wan2.1-T2V-1.3B, then finetuned on 50-frame clips at $[240 \times 360]$ resolution and 10 FPS. The VAE and text encoder are frozen. Training and preprocessing details, including the dataset mixture weights, are provided in the appendix.

4.2 Camera-Controlled Generation

We evaluate the unified model on camera-controlled video generation, testing whether camera-ray conditioning can guide visual synthesis while preserving appearance and scene layout across viewpoint changes.

Benchmarks and Metrics. We evaluate on RealEstate10K, SpatialVID, and DL3DV, covering diverse indoor and outdoor scenes. For each sequence, the context contains the target camera trajectory and either the first RGB frame or the first and last RGB frames. The model then generates the remaining frames from this context, which are evaluated against ground-truth videos using PSNR, SSIM, and LPIPS [46, 60].

Baselines. We compare quantitatively against Aether [67] and SEVA [63], and qualitatively against DFoT [36] and Wan2.1-FLF-14B in a first-last-frame conditioning setting [39]. These baselines represent unified generation and geometry prediction, diffusion-based novel view synthesis, diffusion-forcing-based video completion, and endpoint-conditioned video generation, respectively. In contrast to the task-specific baselines, UGG has the same multimodal architecture for both camera-controlled generation and geometry prediction.

Table 1: **Camera-controlled generation results.** We evaluate generation conditioned on either the first RGB frame (1-view) or the first and last RGB frames (2-view), together with the camera trajectory. We report PSNR (↑), LPIPS (↓), and SSIM (↑) onRealEstate10K, SpatialVID, and DL3DV.

	Method	RealEstate10K			SpatialVID			DL3DV		
		PSNR	LPIPS	SSIM	PSNR	LPIPS	SSIM	PSNR	LPIPS	SSIM
2-view	Aether	20.02	0.196	0.624	13.84	0.489	0.334	13.14	0.399	0.243
	SEVA	25.25	0.081	0.838	15.44	0.344	0.466	17.61	0.211	0.552
	UGG (ours)	25.73	0.061	0.849	16.99	0.288	0.535	16.61	0.239	0.495
1-view	Aether	15.79	0.335	0.469	11.81	0.632	0.283	11.96	0.506	0.240
	SEVA	19.58	0.194	0.703	13.35	0.472	0.343	13.50	0.393	0.335
	UGG (ours)	21.79	0.150	0.779	14.86	0.428	0.405	13.92	0.385	0.346

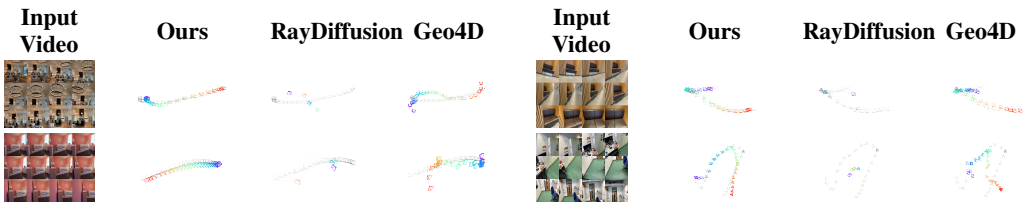


Figure 5: **Qualitative comparison of estimated camera trajectories.** The temporal evolution of each trajectory is encoded by a color gradient, from purple (start) to red (end). The leftmost column shows representative RGB frames from the input video. Compared with Geo4D and RayDiffusion, our predicted trajectories more closely follow the ground-truth trajectory, with better overall structure in challenging cases.

Evaluation results. Camera-controlled generation results are shown in Table 1. Although not trained solely for novel-view synthesis, UGG remains competitive across camera-conditioned generation benchmarks. It achieves the best PSNR, LPIPS, and SSIM on all three datasets in the one-view setting and on RealEstate10K and SpatialVID in the two-view setting; SEVA remains stronger on two-view DL3DV. This shows that ray-map conditioning supports accurate camera control, even within a broader RGB-depth-camera completion model. We visualize generated examples in Figure 4. Wan2.1-FLF does not take camera trajectories as input, and therefore struggles with complex camera motion. We find that UGG also preserves dynamic content better; SEVA is primarily designed for static scenes, and DFoT distorts or remove moving humans.

4.3 Camera Pose Estimation

We evaluate camera pose estimation from RGB observations in dense and sparse settings, conditioning on either the full video or only the first and last frames. Camera-ray tokens are denoised and converted back into camera parameters, testing whether camera geometry can be predicted as a missing modality within the same conditional-completion framework.

Benchmarks and Metrics. We report results on RealEstate10K, Sintel [6], TUM [37], ScanNetV2 [12], and ScanNet++. Following Geo4D [19], we align predicted trajectories to ground truth using Sim(3) Umeyama alignment. In the dense setting, models condition on the full RGB sequence, and we report absolute trajectory error (ATE), relative pose error in translation (RPE_t), and in rotation (RPE_r). ATE measures global trajectory agreement after alignment, while RPE measures local frame-to-frame motion consistency. In the sparse setting, models condition on only the first and last RGB frames, and we evaluate the relative pose between these frames using translation angular error (tAE) and RPE_r. UGG uses shared intrinsics.

Baselines. We compare quantitatively against Geo4D [19], Gen3R [17], and Aether, and qualitatively against RayDiffusion [57]. Geo4D estimates 4D geometry using a generative video prior, while RayDiffusion directly models camera rays through a diffusion denoising process. The geometry-focused baselines are more specialized for pose estimation, whereas UGG predicts camera rays as part of a broader RGB-depth-camera scene representation.

Evaluation results. Table 2 reports camera pose estimation results. In the dense setting, where the full RGB sequence is observed, our model obtains the lowest ATE and RPE_t on all five datasets. For example, ATE is reduced from 0.190 to 0.089 on ScanNet++ compared with Geo4D. These improvements show that predicting camera rays as dense spatial tokens provides accurate global trajectory estimates and locally consistent translations. Figure 5 visualizes representative trajectories. Our predictions better follow the ground-truth motion, while RayDiffusion and Geo4D often produce collapsed paths in challenging settings.

Table 2: **Camera pose estimation results.** For dense evaluation, we report ATE (↓), RPE_t (↓), and RPE_r (↓, degrees). For sparse evaluation, where only the two observed frames are scored, we report translation angular error (tAE, ↓) and RPE_r (↓, degrees).

	Method	RealEstate10K			Sintel			TUM			ScanNetV2			ScanNet++		
		ATE	RPE_t	RPE_r	ATE	RPE_t	RPE_r	ATE	RPE_t	RPE_r	ATE	RPE_t	RPE_r	ATE	RPE_t	RPE_r
Dense	Geo4D	0.026	0.028	0.147	0.160	0.097	0.911	0.052	0.029	1.414	0.074	0.044	1.762	0.190	0.090	5.591
	Gen3R	0.030	0.011	0.141	0.233	0.066	1.538	0.046	0.016	0.943	0.075	0.024	1.018	0.232	0.066	5.760
	Aether	0.018	0.014	0.851	0.180	0.867	1.354	0.041	0.016	1.462	0.075	0.030	1.617	0.225	0.068	5.944
	UGG (ours)	0.010	0.004	0.075	0.107	0.034	0.680	0.017	0.007	1.051	0.039	0.016	0.943	0.089	0.037	3.858

	Method	RealEstate10K		Sintel		TUM		ScanNetV2		ScanNet++	
		tAE	RPE_r	tAE	RPE_r	tAE	RPE_r	tAE	RPE_r	tAE	RPE_r
Sparse	Geo4D	117.079	3.407	108.628	24.160	100.224	11.936	90.812	43.458	82.747	83.019
	Gen3R	44.004	4.197	75.984	24.984	82.875	12.374	79.871	43.481	92.324	82.540
	Aether	39.834	14.969	78.012	33.640	48.436	6.234	70.925	36.720	77.234	81.681
	UGG (ours)	4.287	1.332	28.543	16.138	7.439	1.613	24.363	21.177	34.149	31.594

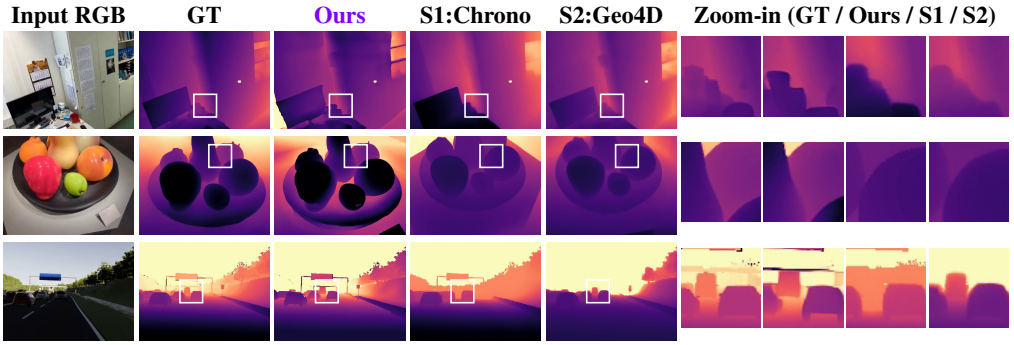


Figure 6: **Depth prediction qualitative comparison.** The white box marks the region shown in the zoom-in columns, ordered as GT, Ours, ChronoDepth, and Geo4D from left to right. Our predictions preserve sharper local structures and object boundaries in the highlighted regions.

The sparse setting is more challenging, since only the first and last RGB frames are provided. Here, the advantage of the generative formulation becomes more pronounced, with UGG obtaining the lowest tAE and RPE_r across all five datasets. This suggests that the model can infer reasonable camera poses from sparse observations, where pose estimation is ambiguous and cannot rely on local matching. By training with diverse conditioning patterns, the model learns a joint RGB-camera prior over plausible trajectories between the observed views.

4.4 Depth Prediction

We evaluate depth prediction from RGB observations in the dense and sparse settings described in the previous section. In both cases, depth tokens are initialized from noise and denoised, testing whether the model learns a useful geometric posterior over scene depth within the same conditional-completion framework.

Benchmarks and Metrics. We evaluate on ScanNet++, 7-Scenes [34], KITTI, Sintel, and Bonn. These datasets cover driving scenes, synthetic video, and indoor RGB-D scans. Following Geo4D [19], we apply video-level scale and shift alignment before evaluation, and report absolute relative error between the true and predicted depth (Abs Rel) and threshold accuracy, whether the predicted depth differs from the ground truth by a less than a factor of 1.25 ($\delta < 1.25$).

Baselines. We compare with Geo4D, ChronoDepth [32], Gen3R, Aether, and Marigold [20]. These baselines include specialized depth and 4D geometry models, as well as a unified model, while ours is trained for RGB-depth-camera completion.

Evaluation results. We report depth prediction results in Table 3. Overall, our model is competitive with specialized depth and 4D geometry baselines across both dense and sparse settings. In the dense setting, it obtains the lowest Abs Rel and highest $\delta < 1.25$ on ScanNet++ among the external baselines. In the sparse setting, where only an image pair is available, our model improves over the baselines on both metrics on 7-Scenes and Bonn. At the same time, Geo4D remains stronger on Sintel and some 7-Scenes metrics, indicating that specialized geometry models still have advantages in some settings. Figure 6 provides qualitative comparisons. The zoom-in crops show that our predictions preserve sharper object boundaries and more coherent local structures in highlighted regions. This suggests that joint RGB-depth training, together with the generative completion objective, helps the model recover fine-scale depth discontinuities while maintaining globally coherent depth maps.

Table 3: **Depth prediction results.** We report Abs Rel (↓) and $\delta < 1.25$ (↑) on ScanNet++, 7-Scenes, KITTI, Sintel, and Bonn in dense and sparse settings.

	Method	ScanNet++		7-Scenes		KITTI		Sintel		Bonn	
		Rel	$\delta_{1.25}$	Rel	$\delta_{1.25}$	Rel	$\delta_{1.25}$	Rel	$\delta_{1.25}$	Rel	$\delta_{1.25}$
Dense	ChronoDepth	0.177	0.724	0.116	0.871	0.120	0.857	0.394	0.530	0.092	0.920
	Geo4D	0.106	0.894	0.080	0.923	0.097	0.923	0.247	0.676	0.057	0.972
	Gen3R	0.157	0.772	0.079	0.937	0.185	0.777	0.414	0.520	0.168	0.805
	Aether	0.166	0.769	0.109	0.898	0.056	0.975	0.351	0.607	0.095	0.911
	Marigold	0.175	0.730	0.094	0.910	0.098	0.910	0.388	0.517	0.082	0.933
	UGG	0.076	0.950	0.078	0.883	0.067	0.967	0.326	0.612	0.056	0.971
	UGG (TTS)	0.075	0.949	0.086	0.916	0.068	0.956	0.297	0.645	0.077	0.955
	UGG (TTS Oracle)	0.067	0.959	0.076	0.923	0.057	0.967	0.263	0.692	0.056	0.973
Sparse	ChronoDepth	0.199	0.700	0.130	0.844	0.089	0.905	0.424	0.487	0.072	0.960
	Geo4D	0.177	0.737	0.098	0.912	0.100	0.897	0.310	0.554	0.067	0.966
	Gen3R	0.238	0.620	0.124	0.867	0.104	0.888	0.347	0.510	0.127	0.847
	Aether	0.269	0.567	0.167	0.772	0.149	0.786	0.487	0.424	0.138	0.849
	Marigold	0.169	0.733	0.106	0.890	0.087	0.921	0.345	0.527	0.071	0.970
	UGG	0.103	0.881	0.087	0.917	0.102	0.914	0.273	0.638	0.051	0.975
	UGG (TTS)	0.104	0.875	0.086	0.919	0.093	0.917	0.237	0.668	0.048	0.978
	UGG (TTS Oracle)	0.056	0.965	0.080	0.923	0.069	0.946	0.183	0.760	0.043	0.979

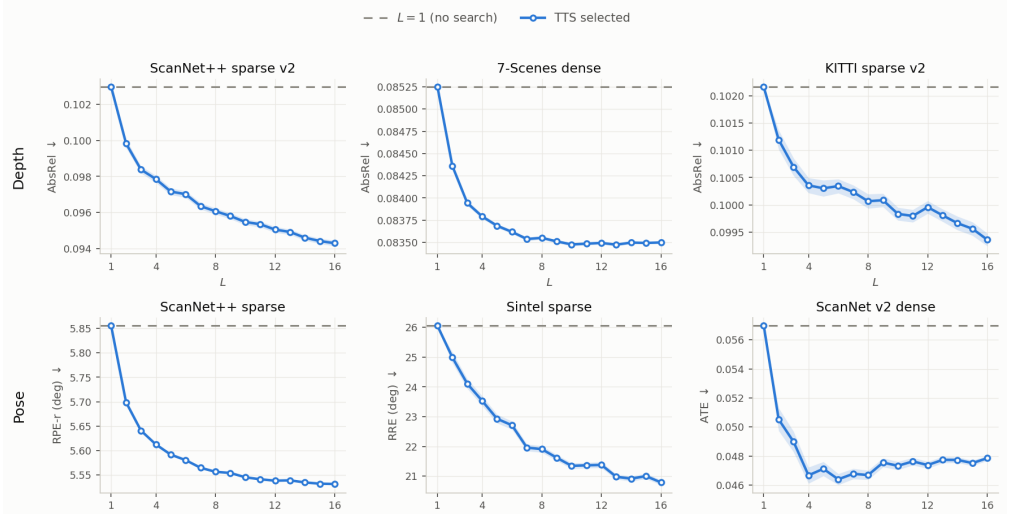


Figure 7: **Test-time search with increasing sample budget.** We sample 16 candidate completions and select the one with the lowest RGB reprojection error. The dashed line shows the single-sample result, and the solid curve shows the selected result as L increases. Search improves several pose and depth settings, showing that the generative model produces useful candidate diversity.

4.5 Test-Time Search on Geometric Prediction

We evaluate whether test-time search (TTS) improves geometric prediction by sampling multiple RGB-depth-camera completions at inference time. For each input, we generate 16 candidates from different noise initializations and select the one with the lowest RGB reprojection error, as described in Section 3.4.

Table 3 reports the selected depth results as *Ours (TTS)*. TTS improves several depth metrics, especially in sparse settings, showing that the generative model can produce useful alternative scene hypotheses. Figure 7 studies the effect of the number of sampled candidates on both pose and depth. Larger search budgets often improve the selected result, but the gains are not uniform because RGB reprojection error is only a proxy for ground-truth geometric accuracy. Overall, these results show that inference-time search is a promising benefit of the generative formulation, while leaving room for stronger search criteria.

4.6 Ablation Studies

Setup. All ablations fine-tune Wan2.1-1.3B on TartanAir [44] and RE10K under a matched budget: 7,500 optimizer steps at batch size 8 (60,000 clips), the same sampled clips, and identical hyperparameters. Each variant changes a single factor. To remove a task, we set its loss weight to zero and replace its input tokens with noise; the remaining loss weights are not renormalized, so every

Table 4: **Ablations.** Each row changes one training task, dataset, architecture, camera representation, or inference setting. We report Abs Rel (↓) and $\delta_{1.25}$ (↑) for depth, ATE (↓) and $\mathrm{RPE_r}$ (↓, degrees) for pose, and PSNR (↑) for generation. RE10K: RealEstate10K. Shading marks the best result within each block.

(a) Depth prediction

	Bonn		KITTI	
Variant	Rel	$\delta_{1.25}$	Rel	$\delta_{1.25}$
Complete architecture	0.135	0.843	0.152	0.791
w/o RGB generation	0.178	0.782	0.322	0.486
w/o RE10K data	0.239	0.651	0.196	0.732
Single transformer	0.129	0.870	0.248	0.640

(b) Camera pose estimation

	ScanNetV2		ScanNet++	
Variant	ATE	RPE_r	ATE	RPE_r
Complete architecture	0.074	2.91	0.193	6.18
w/o depth prediction	0.081	4.30	0.234	7.57
Single transformer	0.083	3.02	0.204	6.60
Dense ray maps	0.071	2.80	0.206	6.09
Global camera tokens	0.103	3.44	0.233	6.61

(c) Camera-controlled generation

	PSNR		
Variant	RE10K	SpatialVID	DL3DV
Full model	20.18	13.67	12.60
w/o pose estimation	13.26	11.87	10.95
Single transformer	19.68	12.75	11.61
Global camera tokens	20.03	12.89	11.64

(d) History-guidance weight

	ScanNetV2		ScanNet++	
w_{hist}	ATE	RPE_r	ATE	RPE_r
0	0.0796	2.46	0.2170	5.77
1	0.0741	2.83	0.2089	6.09
2	0.0740	3.24	0.2091	6.45
4	0.0765	4.22	0.2141	7.70

retained task receives the same supervision as in the complete architecture. For the variant without RGB generation, RGB remains a clean conditioning input.

Cross-task transfer. Removing one task degrades a different one (Table 4a–c). Without RGB generation, KITTI depth Rel rises from 0.152 to 0.322; without depth prediction, ScanNetV2 $\mathrm{RPE_r}$ rises from 2.91° to 4.30°; and without pose estimation, RE10K PSNR drops from 20.18 to 13.26 dB. Joint training therefore helps each task, beyond merely sharing one network.

Heterogeneous data. RE10K provides RGB and camera supervision but no depth labels. Removing it still worsens depth on both Bonn (Rel 0.135 to 0.239) and KITTI (0.152 to 0.196). These results demonstrate that UGG can benefit from heterogeneous training data even when the additional examples do not contain depth annotations.

Architecture and camera representation. Replacing the Mixture-of-Transformers with a single shared transformer lowers pose accuracy, generation quality, and KITTI depth accuracy, although it is slightly better on Bonn. Replacing dense ray maps with global camera tokens degrades pose on both benchmarks and PSNR on all three, supporting image-aligned rays as the interface between camera geometry and visual tokens.

History guidance. Moderate history guidance ($w_{\mathrm{hist}} \in \{1, 2\}$) improves ATE, while rotational RPE increases with guidance strength. High guidance $w_{\mathrm{hist}} = 4$ degrades both metrics. Figure 7 reports sensitivity to the test-time search budget.

5 Conclusion

We present a unified generative-predictive framework for 4D scene understanding, formulating visual generation and geometric prediction as conditional completion over RGB, depth, and camera rays. By changing the clean/noisy pattern over multimodal tokens, a single model supports camera-controlled generation and geometric prediction under dense and sparse observations, while learning from heterogeneous datasets with different modality annotations. Across diverse datasets, our model achieves competitive generation and depth prediction results, strong pose estimation performance, and improved geometric prediction through test-time search. These results suggest that jointly modeling appearance and geometry with generative models is a promising direction toward more general 4D scene understanding.

Limitations. Our current model is not uniformly stronger than specialized methods: task-specific NVS models still perform better on pixel-aligned metrics, and specialized geometry models remain stronger on several indoor depth metrics. Future work could address these gaps with better dataset balancing, stronger geometric supervision, and improved depth-camera consistency during training. Our test-time search also relies on RGB reprojection error, which is useful but not uniformly reliable; learned or multimodal consistency scores may provide stronger search criteria. Finally, training at

larger scale, higher resolution, and with more diverse data could further improve both visual detail and geometric accuracy.

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A Visualizations

Comprehensive visualizations across inference settings are available on our project page, including camera-conditioned generations, camera pose estimation, and depth estimation across datasets.

B Implementation Details

B.1 Datasets and Preprocessing

Datasets. We train a single model on a heterogeneous mixture of real, synthetic, and robotic video datasets: DL3DV-10K [25], RealEstate10K [65], the SpatialVID subset [40], ScanNet++ [53], AgiBot World [5], OmniWorld [66], Virtual KITTI 2 [7], and PointOdyssey [62]. The datasets provide different subsets of RGB, depth, camera pose, and language supervision. This heterogeneity is handled directly by diffusion forcing: unavailable modalities are treated as unobserved targets and their losses are masked out, so datasets with incomplete annotations can still contribute to the shared RGB and cross-modal representation.

Available modalities. For RealEstate10K and the SpatialVID subset, we use the provided RGB frames and camera poses. For DL3DV-10K and ScanNet++, we use RGB frames and obtain depth and camera-pose annotations using the VIPE [16] annotation pipeline. For AgiBot World, we use RGB video supervision only. For synthetic datasets, including OmniWorld, Virtual KITTI 2, and PointOdyssey, we use the available RGB, depth, and camera-pose annotations. We use language conditioning when captions or scene descriptions are available: SpatialVID and AgiBot World provide language descriptions, and we annotate DL3DV-10K and RealEstate10K with scene-level description prompts. For the remaining datasets, we train without text conditioning.

Camera representation. We represent camera pose as a dense ray map aligned with the image plane. For each pixel (u, v), the ray map stores a Plücker ray

$$R_i(u, v) = (\mathbf{d}_i(u, v), \mathbf{m}_i(u, v)), \qquad \mathbf{m}_i(u, v) = \mathbf{o}_i \times \mathbf{d}_i(u, v), \tag{5}$$

where $\mathbf{o}_i$ is the camera center and $\mathbf{d}_i(u, v)$ is the unit viewing direction in the world coordinate frame. The ray direction encodes orientation and intrinsics, while the ray moment encodes the camera center up to the common scene scale. In implementation, ray maps are stored as six-channel tensors with channel order $[\mathbf{d}, \mathbf{m}]$ and are downsampled to the latent spatial resolution.

Cross-dataset normalization. To make supervision from different datasets comparable, we normalize all geometric quantities with a shared clip-level scale. For each video clip, we compute a mutual scale s and use it consistently for camera ray moments and depth. Ray directions are already unit-normalized, while ray moments are divided by s and clamped to the normalized range. For depth, we use a disparity-like parameterization

$$\tilde{D} = \frac{2}{1 + D/s} - 1,$$

where D is the raw metric or reconstructed depth. This maps near depths to values close to 1 and far depths toward -1, yielding a bounded representation that is numerically compatible with RGB and latent diffusion training. With this convention, RGB, depth, and ray-map channels are all represented in approximately the same normalized range $[-1, 1]$.

Missing-modality handling. Each training sample is represented in a common format containing RGB, ray maps, and depth. When a dataset does not provide ray maps or depth, the corresponding tensors are filled with random Gaussian placeholders and marked by modality-availability flags. During noise injection, missing ray-map and depth blocks are forced to the maximum noise level, so the model never receives structured false geometry. During loss computation, unavailable ray-map and depth targets are masked out. This prevents RGB-only or RGB-plus-pose datasets from corrupting geometric supervision while still allowing them to train the shared video-generation backbone.

B.2 Model Architecture

Base model. We initialize from Wan2.1-T2V-1.3B and extend its diffusion transformer into a multimodal Mixture-of-Transformers backbone. The pretrained video VAE and text encoder are kept frozen, and we finetune only the diffusion transformer. The model jointly processes four latent streams,

$$[z^I,\ z^{R_d},\ z^{R_m},\ z^D], \tag{6}$$

corresponding to RGB video, ray directions, ray moments, and depth, respectively. The two ray streams use separate latent blocks but share the same ray-stream transformer weights.

Tokenization. All clips contain 50 frames resized to 240×320. RGB frames are encoded by the frozen Wan VAE into latent tensors with 16 channels and spatial size 30×40. Depth maps are first repeated to three channels and are encoded by the same frozen VAE. Ray maps bypass the VAE: we split each ray map into direction and moment fields and pack them directly into latent tensors at the same spatial resolution as the VAE latents. For the default 50-frame setting, the temporal latent length per modality is $T_{\text{lat}} = 14$. Thus, each modality stream has shape $[B, 16, 14, 30, 40]$, and the concatenated four-stream latent has shape $[B, 16, 56, 30, 40]$. We use a 3D patch size of $[1, 2, 2]$, giving a per-modality patch grid of $14 \times 15 \times 20$, or 4200 tokens per modality and 16800 tokens in total.

Ray latent packing. For ray directions and ray moments, four consecutive 3-channel ray frames are packed into the first 12 channels of a 16-channel latent tensor, with the remaining channels padded by zeros. In the first–last-frame setting, the first latent slot repeats the first frame, the middle slots pack groups of interior frames, and the final latent slot repeats the last frame. This packing keeps ray-map tokens spatially and temporally aligned with RGB and depth VAE tokens while avoiding unnecessary VAE compression of camera geometry.

Mixture-of-Transformers backbone. The transformer uses modality-specific input projections and feed-forward layers, together with joint multimodal self-attention. Concretely, we use separate 3D patch embeddings for RGB, ray maps, and depth. Queries, keys, values, and output projections are stream-specific for the RGB, ray, and depth streams, but the resulting tokens are concatenated and attend to each other in a single multimodal attention operation. Each transformer block also contains stream-specific normalization layers and feed-forward networks. This design preserves modality-specific feature statistics while allowing RGB, depth, and camera geometry to exchange information through shared attention.

Model dimensions. The diffusion transformer follows the Wan2.1-T2V-1.3B configuration: hidden dimension 1536, feed-forward dimension 8960, 12 attention heads, and 30 transformer layers. The VAE latent channel dimension is 16, the VAE stride is $[4, 8, 8]$ in time, height, and width, and the transformer patch size is $[1, 2, 2]$. Text conditioning uses the Wan text-conditioning path with prompt embeddings of dimension 4096 and a maximum length of 512 tokens.

Initialization of new modality branches. When initializing from the RGB-only Wan2.1 checkpoint, all compatible transformer weights are copied from the pretrained model. The RGB patch embedding and output-head weights are used to initialize the new ray and depth patch embeddings and heads. Attention, cross-attention, normalization, feed-forward, time-embedding, and text-conditioning weights are copied into the corresponding modality streams when shapes are compatible. This initialization preserves the pretrained video prior while allowing the new modality branches to specialize during finetuning.

B.3 Training Objective and Diffusion Forcing

Continuous flow matching. We train with continuous flow matching. For each latent token, we sample a raw diffusion time $t \in [0, 1]$ and apply the shifted schedule

$$t' = \frac{\alpha t}{1 + (\alpha - 1)t}, \qquad \alpha = 3.0. \tag{7}$$

Given a clean latent z_0 and Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$, the noisy input is

$$z_{t'} = (1 - t')z_0 + t'\epsilon, \tag{8}$$

and the model predicts the flow target

$$v^\star = \epsilon - z_0. \tag{9}$$

The loss is the mean-squared error between the predicted and target flow, computed separately for RGB, ray direction, ray moment, and depth:

$$\mathcal{L} = \frac{1}{4}\left(\mathcal{L}_I + \mathcal{L}_{R_d} + \mathcal{L}_{R_m} + \mathcal{L}_D\right). \tag{10}$$

For samples without ray-map or depth supervision, the corresponding losses are masked out.

Diffusion-forcing modes. We specify training tasks by assigning different diffusion levels to different modality-frame groups. Clean or low-noise tokens serve as conditioning context, while high-noise tokens are prediction targets. We use five conditioning patterns:

1. First RGB latent as history; remaining RGB, ray maps, and depth as targets;
2. All RGB latents as history; ray maps and depth as targets;
3. First and last RGB latents together with all ray-map latents as history; middle RGB and depth as targets;
4. First RGB latent together with all ray-map latents as history; remaining RGB and depth as targets;
5. First RGB latent together with all depth latents as history; remaining RGB and ray maps as targets.

These patterns correspond to image-conditioned scene completion, RGB-to-geometry prediction, camera-conditioned video generation, camera-conditioned extrapolation, and depth-conditioned generation/prediction. We sample these modes during training and restrict them according to the modalities available in each dataset. For example, depth-conditioned modes are used only for datasets with depth supervision, and ray-conditioned modes are used only for datasets with camera-pose or ray-map supervision.

History noise. For history tokens, we use clean context with probability 0.5. Otherwise, history tokens receive a random lower noise level. Target tokens receive independently sampled future noise levels. This teaches the model to use both perfectly clean context and imperfect context, which improves robustness when conditioning on generated or partially corrupted histories.

Optimization. We train on 50-frame clips at 240×320 resolution with a reference frame rate of 10 FPS. Optimization uses AdamW with learning rate 1×10^{-5}, betas $(0.9, 0.95)$, weight decay 0.05, gradient clipping at 5.0, and a constant learning-rate schedule with 1000 warmup steps. Training is performed with mixed precision on 4 NVIDIA H200 GPUs for approximately $30{,}000$ optimizer steps. We use batch size 1 per GPU and gradient accumulation of 4, giving an effective batch size of 16. We use gradient checkpointing for memory efficiency.

Dataset sampling weights. The training mixture uses relative dataset weights. In our final configuration, the relative weights are 0.5 for RealEstate10K, 5.0 for DL3DV-10K, 1.0 for SpatialVID, 1.5 for OmniWorld, 30.0 for ScanNet++, 50.0 for Virtual KITTI 2, 50.0 for PointOdyssey, and 0.04 for AgiBot World. The weights are multiplied by dataset size and normalized by the mixture loader.

B.4 Inference

Sampling. At inference time, the same clean/noisy pattern specifies the task. Observed RGB, depth, or ray-map tokens are clamped to their clean values at every denoising step and assigned zero diffusion level. Unknown tokens are initialized from Gaussian noise and progressively denoised. We use UniPC sampling with 40 denoising steps and the same shifted time schedule used during training. Under this setting, generating one 50-frame clip takes approximately one minute on a single NVIDIA H100 GPU.

History guidance. We use history guidance to strengthen consistency with observed context. For a set of observed history tokens H, we compute a history-conditioned flow $\hat{v}_H$ with H clamped to the observed values and a history-dropped flow $\hat{v}_\emptyset$ in which the same context is replaced by noise. The guided flow is

$$\hat{v}_{\mathrm{HG}} = \hat{v}_H + w_{\mathrm{hist}} \left(\hat{v}_H - \hat{v}_\emptyset \right). \tag{11}$$

We set $w_{\mathrm{hist}} = 2.0$ in our default configuration.

Decoding. After denoising, RGB and depth latents are decoded with the frozen Wan VAE. Ray-direction and ray-moment latents are unpacked into dense ray maps. For camera-pose evaluation, the predicted ray maps are converted back to camera parameters using the same ray-map-to-camera recovery procedure across all datasets.

Inference-time search. For test-time search, we sample multiple candidate completions from different random seeds while clamping the same observed context. Each candidate produces RGB, depth, and ray-map predictions. We score candidates by cross-frame RGB reprojection error using

Asset	Type	Use in this work	License / access terms
DL3DV-10K [25]	Dataset	Training and evaluation for RGB video, camera pose, and depth/camera-conditioned generation.	DL3DV-10K Terms of Use and Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0); non-commercial research and educational use.
RealEstate10K [65]	Dataset	Training and evaluation for camera trajectories, camera-controlled generation, and pose prediction.	Creative Commons Attribution 4.0 International (CC BY 4.0), following the official RealEstate10K release.
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ScanNet++ [53]	Dataset	Training and evaluation on indoor RGB-D scenes with camera and geometry annotations.	ScanNet++ Terms of Use; non-commercial research and educational use only, with restricted redistribution.
AgiBot World [5]	Dataset	Training on robotic manipulation videos with multimodal embodied-scene observations.	Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International (CC BY-NC-SA 4.0).
OmniWorld [66]	Dataset	Training on multi-domain 4D world-modeling data.	Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International (CC BY-NC-SA 4.0), with additional restrictions and notices for third-party game content.
Virtual KITTI 2 [7]	Dataset	Training and evaluation for synthetic driving scenes with RGB, depth, camera, and motion annotations.	Non-commercial use only; Creative Commons Attribution-NonCommercial-ShareAlike 3.0 (CC BY-NC-SA 3.0).
PointOdyssey [62]	Dataset	Training on synthetic dynamic scenes with camera, depth, segmentation, and long-term point/geometry annotations.	Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International (CC BY-NC-SA 4.0) for the released dataset.
7-Scenes [34]	Dataset	Evaluation-only dataset for indoor RGB-D depth prediction.	Microsoft Research 7-Scenes license agreement; non-commercial use only.
Wan2.1-T2V-1.3B [39]	Pretrained model	Initialization for the video diffusion backbone; we finetune the diffusion transformer while using the pretrained VAE and text encoder components.	Apache License 2.0, following the official Wan2.1-T2V-1.3B release.

Table 5: Summary of existing datasets and pretrained model assets used in this work. We cite the original creators of each asset and use each asset according to its official license or access terms. We do not redistribute third-party datasets or restricted source media.

their own predicted depth and camera geometry, and select the candidate with the lowest reprojection error. This implements the inference-time search procedure described in Sec. 3.4.

C Existing Assets and Licenses

We use existing public datasets and pretrained model assets. Table 5 summarizes the role of each asset in this work and the corresponding license or access terms. All datasets are used for academic research purposes only. We do not redistribute the original datasets, videos, game assets, or other third-party data. Any released code will provide dataset preprocessing and evaluation scripts, while requiring users to obtain the underlying datasets from their official sources and comply with the corresponding terms of use. If model checkpoints are released in the future, they will be released only under terms compatible with the upstream data and model licenses.

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